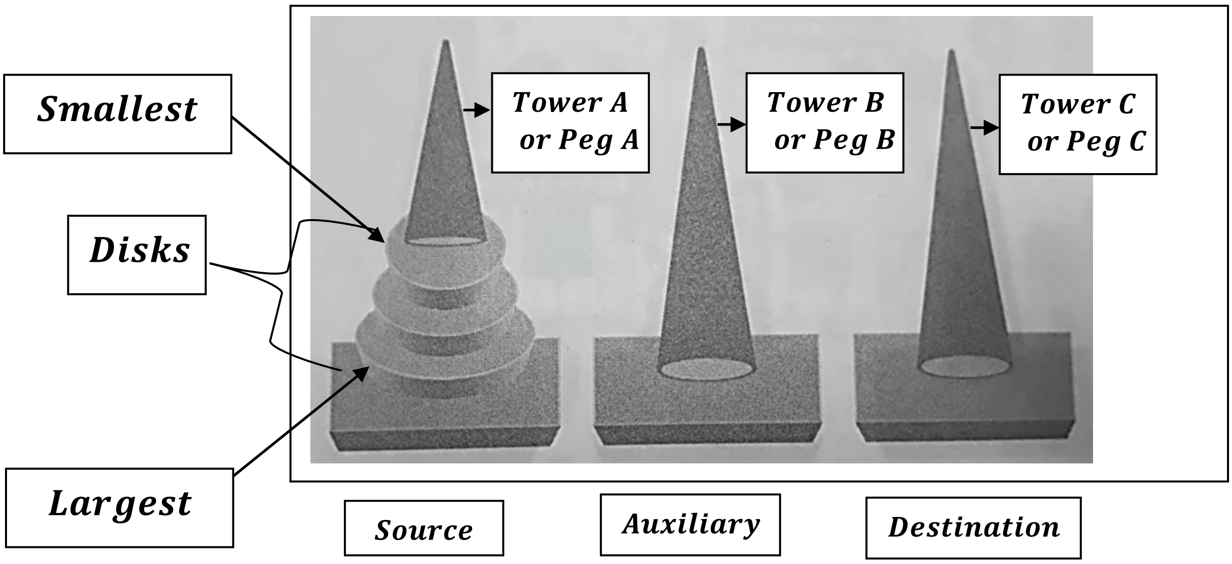
$How\ TOH(Tower\ of\ Hanoi)\ Works?$

$Solve\ the\ TOH(Tower\ Of\ Hanoi) with\ n = 3\ , where\ n\ is\ the\ number\ of\ disks.$



$Challenge$:

1. $One\ disk\ should\ move\ to\ a\ tower\ one\ at\ a\ time.$

2. $Not\ allowed\ to\ move\ a\ larger\ disk\ on\ top\ of\ a\ smaller\ disk.$

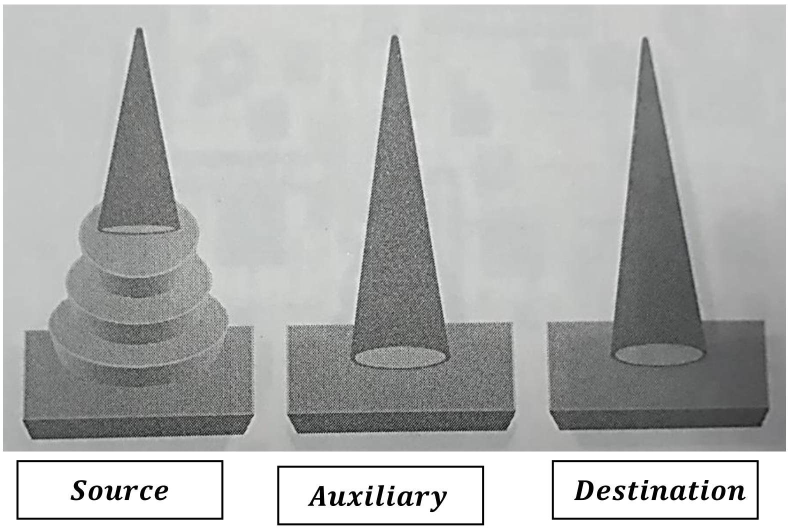
3. $Disk\ should\ get\ arranged\ on\ Destination\ Tower\ from\ source\ tower\ with\ the\ help\ of\ auxiliary\ tower.$

Where ,

- ***Source Tower is Tower A***.
- ***Destination Tower is Tower C***.
- ***Auxiliary Tower is Tower B***.

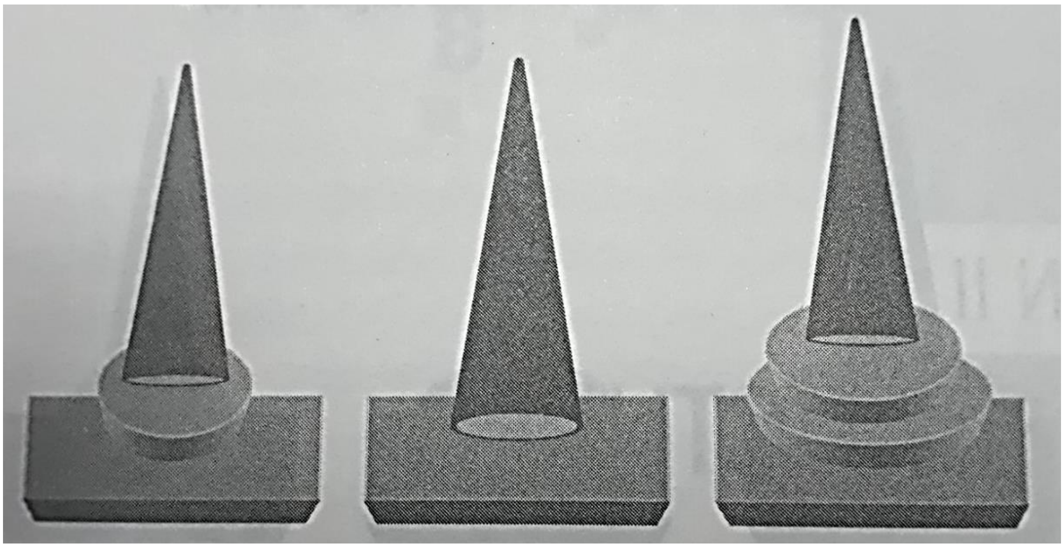
<u>Solution</u>

Initializationed Disks:



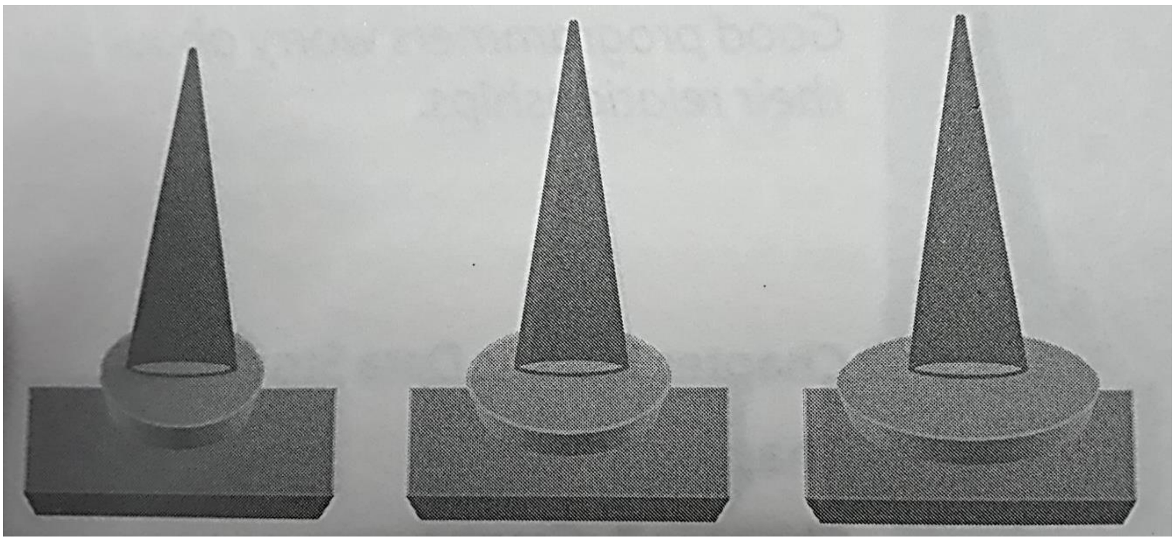
There are three disk in the first peg.

1st the smallest disk moved to third tower



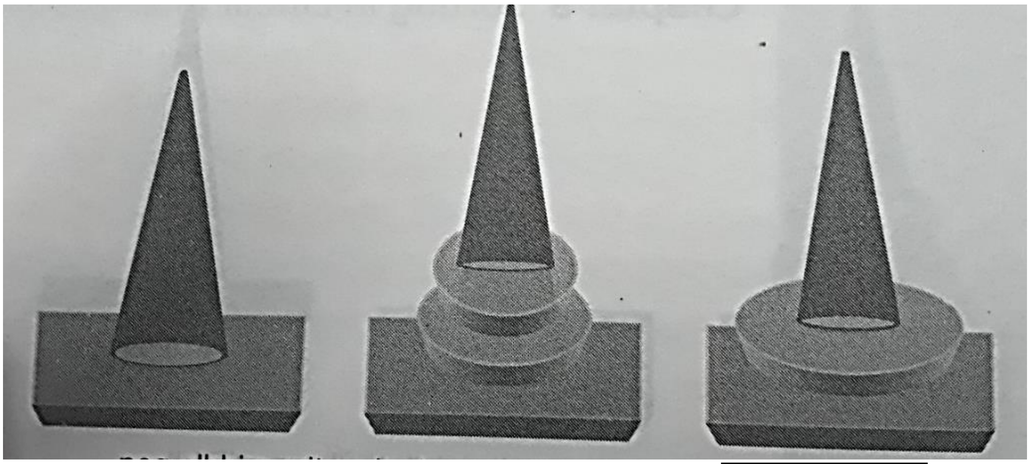
Source Auxiliary Destination

2nd the second larger disk or medium sized disk moved to second tower.



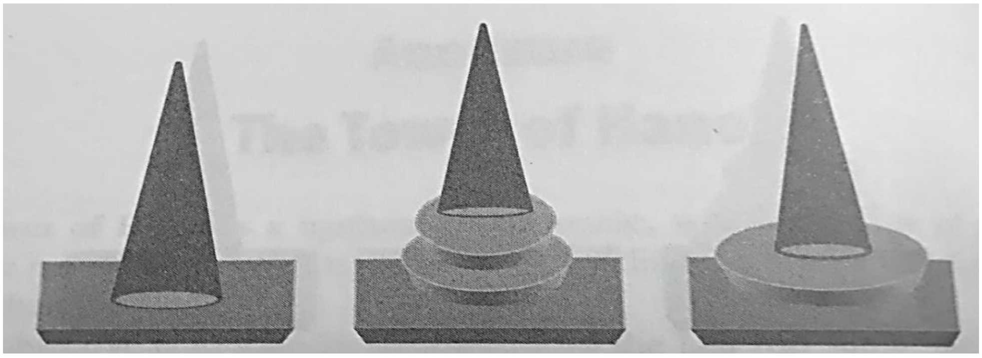
Source Auxiliary Destination

3rd the smallest disk from Tower C moved to Tower B and put on the larger(medium sized) disk.



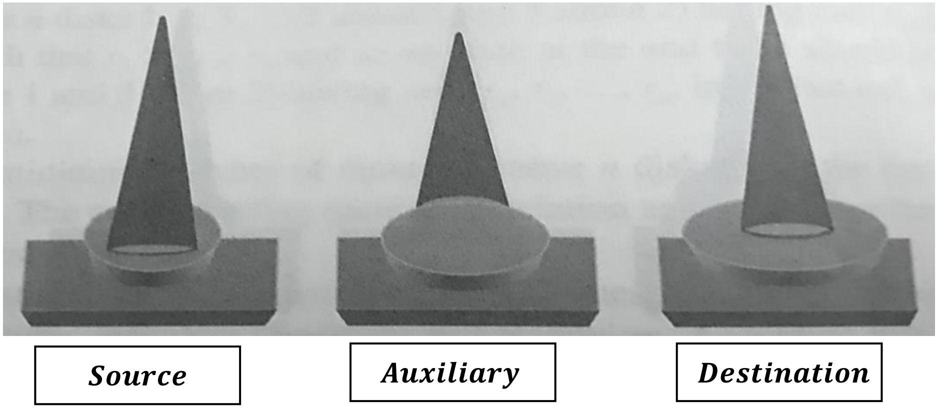
Source **Auxiliary** **Destination**

4rth – The largest disk from Tower A moved to destination Tower C .

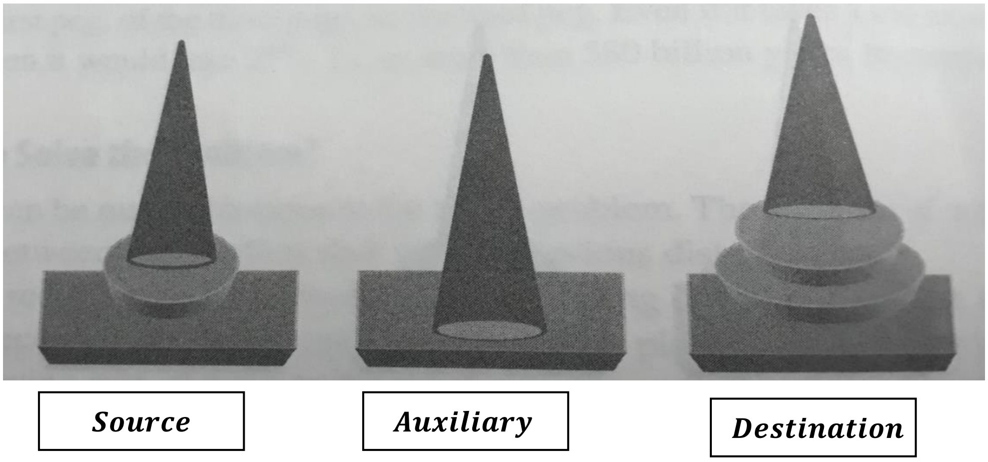


Source **Auxiliary** **Destination**

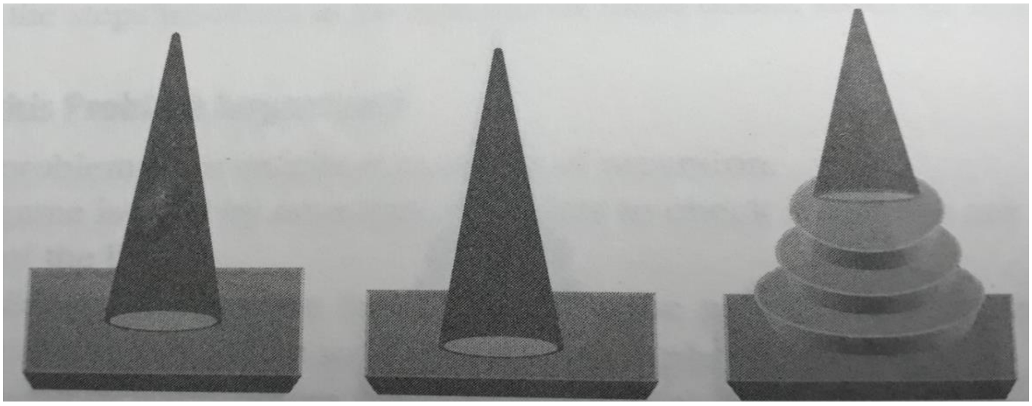
5th – The smallest disk from Tower B moved to source tower Tower A .



6th – The medium sized disk is put on the largest disk and moved from Auxiliary Tower B to destination tower C.



7th – The smallest disk is put on the medium sized disk finishing up the formation at destination tower C.



Source ***Auxiliary*** ***Destination***

$\therefore$ ***How many steps taken? That is 7 steps.***

i.e. $\mathbf{2^3 - 1}$ ***where 3 is the number of disk. Hence we can say no. of steps taken in tower of Tower of Hanoi is*** $\mathbf{2^n - 1}$ ***where n is no. of disks.***

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